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Oral appliance for the treatment of sleep apnea

Abstract

An oral appliance may include a mouthpiece configured to be positioned in an oral cavity of a user and an electronic assembly coupled to the mouthpiece. The electronic assembly may include an oxygen sensor oriented toward soft tissue of the user's mouth to determine an oxygen saturation level of the user. The electronic assembly may be positioned within an appendage extending away from a wall of the mouthpiece.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of and claims priority to U.S. patent application Ser. No. 18/209,298 filed Jun. 13, 2023. This application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/497,605 filed on Apr. 21, 2023, the entire contents of each of which is incorporated by reference herein.

BACKGROUND

(1) Sleep apnea is a common medical condition during which a person experiences one or more pauses in breathing, and in some instances, experiences shallow breaths during sleep. While there are several types of sleep apnea, the most common type is obstructive sleep apnea. In this medical condition, one or more of the person's throat muscles relax during sleep causing surrounding tissues in the posterior portions of the mouth, nose and throat to collapse, thereby creating a pharyngeal obstruction that can block the upper airway. Persons suffering from obstructive sleep apnea have inadequate oxygen exchange during sleep, which can lead to daytime fatigue, lack of concentration and mood changes. Left untreated, obstructive sleep apnea can have a significant impact on a person's health, often leading to cardiovascular, stroke and metabolic disorders.

(2) To reduce this risk, various nonsurgical approaches have been employed. One such nonsurgical approach includes using standardized oral appliances to incrementally advance and/or protrude the mandible (lower jaw) relative to the maxilla (upper jaw). These standardized oral appliances, commonly referred to as a mandibular advancement device, ("MAD"), typically include upper and lower dental trays, whereby the lower dental tray is designed to advance the mandible, and hence, move the tongue forward to increase the space in the posterior part of the throat and the oropharynx, which in turn may serve to increase the flow of air during sleep. The distance (degree of advancement) required to protrude and/or reposition the mandible may be, at least in part, dependent on the severity of the individual's obstructive sleep apnea, as well as psychological variables among the users. A disadvantage of using these standard oral appliances is that they may not sufficiently provide for and/or address individualized anatomical variances, such as difference in dental arches, dentition alignment and/or jaw flexibility. Another disadvantage is that in instances where the degree of advancement is excessive, the appliance may lead to long-term temporomandibular joint ("TMJ") disorders, muscular aggravation, dentition discomfort and/or myofascial disorders. As a result, use of these standard appliances has an approximate compliance rate of 75% over a 2-year period. For a detailed study of compliance with use of MAD, see *Non-CPAP therapies in obstructive sleep apnoea: mandibular advancement device therapy*, see Eur Respir J 2012; 39:1241-1247, which is incorporated by reference in its entirety. Thus, such oral appliances may not treat obstructive sleep apnea in a manner that prevents and/or limits impacts on a person's health.

(3) In view of the disadvantages associated with currently available methods and devices for treating obstructive sleep apnea, there is a need for a device and method that treats obstructive sleep apnea while storing patient behavior and/or medical data relating to a user's oxygen saturation, breathing pattern, snoring pattern and/or clenching/grinding behaviors, that can assist medical providers in the design, improvement and/or modification of specialized treatment measures for individual patients. Further, there is a need for a device and method that treats obstructive sleep apnea in a single removable oral appliance and prevents and/or limits long-term TMJ disorders, muscular aggravation and/or myofascial disorders that may occur with continued use of currently available appliances.

BRIEF DESCRIPTION

(4) According to an aspect of this disclosure, an exemplary embodiment of an oral appliance may include a mouthpiece that is configured to be positioned in an oral cavity of a user. The mouthpiece may include an anterior wall configured to cover facial surfaces of the user's dentition, a posterior wall configured to cover lingual surfaces of the user's dentition, and a transverse wall extending between the anterior wall and the posterior wall. An electronic assembly may be coupled to the mouthpiece and the electronic assembly may include an oxygen sensor coupled to the anterior wall and configured to be oriented toward soft tissue of the user's mouth to determine an oxygen saturation level of the user.

(5) According to an aspect of this disclosure, an exemplary embodiment of an oral appliance may include a mouthpiece configured to be positioned in an oral cavity of a user. The mouthpiece may include an anterior wall configured to cover facial surfaces of the user's dentition, a posterior wall configured to cover lingual surfaces of the user's dentition, and a transverse wall extending between the anterior wall and the posterior wall. An electronic assembly may be coupled to the mouthpiece. The electronic assembly may include an oxygen sensor coupled to the anterior wall and configured to be oriented toward soft tissue of the user's mouth to determine an oxygen saturation level of the user and collect oxygen saturation (SpO2) and photoplethysmography (PPG) data.

(6) According to an aspect of this disclosure, an exemplary embodiment of a oral appliance may include a mouthpiece configured to be positioned in an oral cavity of a user. The mouthpiece may include an anterior wall configured to cover facial surfaces of the user's dentition, a posterior wall configured to cover lingual surfaces of the user's dentition, and a transverse wall extending between the anterior wall and the posterior wall. An appendage may extend away from the anterior wall in a direction towards soft tissue of the user's mouth and define a space within the appendage. An electronic assembly including an oxygen sensor and may be positioned at least partially within the space within the appendage, and the appendage and electronic assembly may together be configured for orienting the oxygen sensor toward the soft tissue of the user's mouth.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) A more particular description will be rendered by reference to specific embodiments thereof that are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments thereof and are not therefore to be considered to be limiting of its scope, exemplary embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1 is a perspective view of an exemplary oral appliance in accordance with aspects of the present disclosure;

(3) FIG. 2 is a plan view of the oral appliance of FIG. 1

(4) FIG. 3 is a plan view of the oral appliance of FIG. 1, with a cross-section taken along a horizontal plane of a housing assembly of the oral appliance;

(5) FIG. 4 is a posterior view, with a posterior housing half removed, illustrating internal electrical components of a housing assembly of the oral appliance of FIG. 1;

(6) FIG. 5 is an anterior view, with an anterior housing half removed, illustrating the internal electronic components of the housing assembly of FIG. 4;

(7) FIG. 6 is a side, cutaway view of the housing assembly of the oral appliance of FIG. 1;

(8) FIG. 7 is a plan view illustrating the internal electronic components of FIG. 6;

(9) FIG. 8A is an anterior view of the internal electronic components of FIG. 6;

(10) FIG. 8B is a top view of the internal electronic components;

(11) FIG. 8C is a posterior view of the internal electronic components;

(12) FIG. 9 is a schematic illustration of components of an oxygen sensor positioned between

isolation barriers of the oral appliance of FIG. 1;

(13) FIG. 9A is an oxygen sensor according to an exemplary embodiment;

(14) FIG. 10 is a side, perspective view illustrating another aspect of an oral appliance including a top mouthpiece, a bottom mouthpiece, and a strut assembly;

(15) FIG. 11 is a plan view illustrating internal components of the top mouthpiece of the oral appliance of FIG. 10;

(16) FIG. 12A is a plan view illustrating a housing of the top mouthpiece of FIG. 11;

(17) FIG. 12B is a side, perspective view illustrating the housing of FIG. 12A;

(18) FIG. 12C is a side, perspective view illustrating a cover of the top mouthpiece of FIG. 10;

(19) FIG. 13 is a perspective view illustrating the top mouthpiece of the oral appliance of FIG. 10 positioned on a user's upper dentition;

(20) FIG. 14 is a side, perspective view illustrating the oral appliance of FIG. 10 positioned on the upper dentition of the user;

(21) FIG. 15 is a chart that depicts data relating to snoring and normal breathing sounds, collected by a sound measuring device, according to an aspect of the present disclosure;

(22) FIG. 16 is a chart that depicts data relating to snoring and normal breathing sounds, collected by a sound measuring device, according to an aspect of the present disclosure;

(23) FIG. 17 is a chart that depicts data relating to normal breathing sound, collected by a sound measuring device, according to an aspect of the present disclosure;

(24) FIG. 18 is a chart that depicts data relating to normal breathing sound, collected by a sound measuring device, according to an aspect of the present disclosure;

(25) FIG. 19 is a chart that depicts data relating to snoring and normal breathing sounds, collected by a sound measuring device, according to an aspect of the present disclosure;

(26) FIG. 20 is a chart that depicts data relating to normal breathing sound, collected by a sound measuring device, according to an aspect of the present disclosure;

(27) FIG. 21 is a chart that depicts data relating to snoring sounds, collected by a sound measuring device, according to an aspect of the present disclosure;

(28) FIG. 22 is a chart that depicts data relating to snoring sounds, collected by a sound measuring device, according to an aspect of the present disclosure;

(29) FIG. 23 is a chart that depicts data relating to normal breathing sound, collected by a sound measuring device, according to an aspect of the present disclosure;

(30) FIG. 24A are two charts illustrating data collected by additional sensors such as an accelerometer and a gyroscope of an oral appliance of the present disclosure with a patient with an open mouth and an inertial measuring unit (IMU) intra-orally coupled to the oral appliance;

(31) FIG. 24B are two charts illustrating data collected by an accelerometer and a gyroscope of an oral appliance of the present disclosure with a patient with an open mouth and an IMU extra-orally coupled to the patient's chin;

(32) FIG. 25A are two charts illustrating data collected by an accelerometer and a gyroscope of an oral appliance of the present disclosure with a patient grinding their teeth in a side-to-side motion and an IMU intra-orally coupled to the oral appliance;

(33) FIG. 25B are two charts illustrating data collected by an accelerometer and a gyroscope of an oral appliance of the present disclosure with a patient grinding their teeth in a side-to-side motion and an IMU extra-orally coupled to the patient's chin;

(34) FIG. 26A are two charts illustrating data collected by an accelerometer and a gyroscope of an oral appliance of the present disclosure with a patient swallowing and an IMU intra-orally coupled to the oral appliance;

(35) FIG. 26B are two charts illustrating data collected by an accelerometer and a gyroscope of an oral appliance of the present disclosure with a patient swallowing and an IMU extra-orally coupled to the patient's chin;

(36) FIG. 27 is a perspective view illustrating yet another embodiment of an oral appliance

including a sensor assembly;

(37) FIG. **28** is a partial, perspective view illustrating the sensor assembly of FIG. **27** with a charging coil thereof removed;

(38) FIG. **29** is a partial, perspective view illustrating the sensor assembly of FIG. **27** with the charging coil and a spacer thereof removed;

(39) FIG. **30** is another partial, perspective view illustrating a printed circuit board and a sensor of the sensor assembly of FIG. **29** positioned in the housing;

(40) FIG. **31** is a bottom, perspective view illustrating an oral appliance including an electromyogram sensor according to another embodiment of the present disclosure;

(41) FIG. **32** is a chart illustrating electromyogram data collected during tongue muscle stimulation and while a user's eyes are moved between opened and closed configurations; and

(42) FIG. **33** is a chart illustrating the difference between electromyogram data from tongue muscle activation versus electromyogram data in the absence of tongue muscle activation.

(43) Various features, aspects, and advantages of the embodiments will become more apparent from the following detailed description, along with the accompanying figures in which like numerals represent like components throughout the figures and text. The various described features are not necessarily drawn to scale but are drawn to emphasize specific features relevant to some embodiments.

DETAILED DESCRIPTION

(44) Reference will now be made in detail to various embodiments. Each example is provided by way of explanation and is not meant as a limitation and does not constitute a definition of all possible embodiments.

(45) For purposes of illustrating features of the embodiments, embodiments will now be introduced and referenced throughout the disclosure. Those skilled in the art will recognize that this example is illustrative and not limiting and is provided purely for explanatory purposes.

(46) As used herein, “coupled” may refer to any manner of assembling or connecting two components together in any suitable manner, such as, by way of example only: attaching directly or indirectly (e.g., attached to a surface), disposing on, disposing within, disposing substantially within, formed with, embedding within, embedded substantially within, etc. “Coupled” may further include fixedly attaching two components (such as by using a screw or embedding a first component into a second component during a manufacturing process), but does not so require. That is, two components may be coupled temporarily simply by being in physical contact with one another.

(47) In an embodiment, and with particular reference to FIGS. **1-6**, an oral appliance **10** for treatment of sleep apnea in a user is provided. The oral appliance **10** generally includes a mouthpiece **20** and a housing assembly **30** coupled to the mouthpiece **20**. In an embodiment, the mouthpiece **20** is “customizable,” that is, customized to the individual user's mouth in such a manner that it provides for a comfortable fit over and around surfaces of the user's hard (teeth/dentition) and/or soft tissues (general mouth structure, including gingiva). When customized, the mouthpiece **20** may fit over temporary, permanent, primary natural and/or artificial lower or upper dentition of adult and/or child users. The mouthpiece **20** may be configured to receive a removable denture of the user. According to an aspect, the mouthpiece **20** is fabricated over the upper jaw or lower jaw with partial or complete absence of dentition. When customized, the mouthpiece **20** can be formed of any suitable self-conforming material that may be adaptable to variances and/or changes in mouth structure, or through use of a dental impression of the individual user's dentition, as would be understood by a person having ordinary skill in the art. In other words, a mandibular impression and/or a dental impression can be taken, whereby a negative imprint of the user's hard and/or soft tissues are used to create a positive reproduction (or cast) customized for the user. In aspects, the mandibular impression and/or dental impression may be a digital dental impression taken with 3D scanning technology or other suitable technologies.

(48) The types of materials selected to form the mouthpiece **20** would be known to one of ordinary skill in the art and includes polymers, thermoplastics, acrylics, silicone, rubber, metal wires or any other material that can be used to form the mouthpiece **20** conformed to the user's dentition. In an embodiment, the materials are medical-grade, latex-free, BPA-free and any other material known to minimize patient health risks. According to an aspect, the mouthpiece **20** may be formed from the impression made in a suitable impression material, such as, for example, alginate, polysulfide, polyvinyl siloxane, silicone, or the like. The mouthpiece material may also be selected, particularly from polymers, for its ability to have a pharmaceutical compound incorporated within the structural matrix.

(49) The mouthpiece **20** includes an anterior wall **22** configured to cover facial surfaces of a user's dentition, a posterior wall **24** configured to cover lingual surfaces of the user's dentition, and a transverse wall **26** interconnecting the anterior wall **22** and the posterior wall **24** and configured to cover an occlusal surface of the user's dentition. The mouthpiece **20** defines a central channel **28** bounded by the anterior wall **22**, the posterior wall **24**, and the transverse wall **26**. The central channel **28** is configured to be positioned over one or more of the user's dentition such that the mouthpiece **20** is secured thereon. When the mouthpiece **20** is in use, the central channel **28** may receive the user's dentition and may extend over and/or cover occlusal or bite surfaces of the user's dentition. The posterior wall **24** of the mouthpiece **20** extends between the user's dentition and the user's tongue. In an embodiment, the anterior wall **22** of the mouthpiece **20** is configured to extend between the user's dentition and the user's cheek.

(50) The mouthpiece **20** may include a socket **32** coupled to the anterior wall **22** of the mouthpiece **20**. The socket **32** is configured to couple to a strut assembly, such as, for example, one or more of the strut assemblies disclosed in U.S. patent application Ser. No. 17/737,470 filed May 5, 2022, the entire contents of which are incorporated by reference herein. The oral appliance **10**, in combination with the strut assemblies, may be configured to position the bottom jaw of a user in a direction that is forward from the natural position of the user's jaw to aid in the treatment of sleep apnea in the user. The best jaw position for the user may be determined by a dentist and prescribed to a lab that is tasked with fabricating the oral appliance **10**.

(51) The housing assembly **30** of the oral appliance **10** generally includes a housing **34** and a printed circuit board assembly **40** (FIGS. 2 and 3) positioned within the housing **34**. The housing **34** has an anterior portion **36**, and a posterior portion **38** coupled to the anterior wall **22** of the mouthpiece **20**. The anterior portion **36** and the posterior portion **38** may be monolithically formed with one another to form the housing **34**, and in other aspects the anterior portion **36** and the posterior portion **38** may be separate components that are otherwise attached to one another. The posterior portion **38** of the housing **34** conforms to and is coupled to the anterior wall **22** of the mouthpiece **20** using any suitable fastening engagement, such as, for example, adhesives, fasteners, heat treatment, or the like. In other aspects, the posterior portion **38** of the housing **34** may be monolithically formed with the mouthpiece **20**.

(52) The posterior portion **38** protrudes anteriorly away from the mouthpiece **20** and perpendicularly away from a gingival edge **27** of the mouthpiece **20**. The posterior portion **38** may be fabricated from a light-transmissive material, such as, for example, polycarbonate, acrylic, polyethylene terephthalate, amorphous copolyester (modified forms of polyester, such as combinations of diacids and diols), polyvinyl chloride, liquid silicone rubber, polyethylene, styrene methyl methacrylate, or the like. In aspects and as illustrated in FIG. 3, only a portion of the posterior portion **38** may be light-transmissive, such as, for example, an optical window **42** that is positioned at a midline of the posterior portion **38**. With reference to FIG. 6, the optical window **42** is monolithically formed with the posterior portion **38** and may also protrude posteriorly from the posterior portion **38** to define a recess **44** therein for receiving an optical assembly **50** therein, as will be described in more detail below. As shown herein, the material forming the posterior portion **38** differs from the material forming the anterior portion **36**.

(53) With particular reference to FIGS. 8A-C, the housing **34** defines an inner chamber **46** therein that extends along a length of the housing **34** for housing the printed circuit board assembly **40** therein. The printed circuit board assembly **40** includes a printed circuit board **48**, the optical assembly **50**, and in some aspects, a microprocessor (not labeled) and other suitable electronic components. For example, the printed circuit board assembly **40** may further include a wireless power charging coil **52** positioned at a first end portion of the printed circuit board **48** and one or more batteries **54**, such as, for example, a lithium-polymer battery (LiPo) rechargeable batteries, positioned at the first end portion and a second end portion of the printed circuit board **48**.

(54) The optical assembly **50** is positioned adjacent the optical window **42** of the housing **34** and includes an optical housing **56** supported on a posterior side of the printed circuit board **48** and an oxygen sensor **60** supported in the optical housing **56** and electrically coupled to the microprocessor of the printed circuit board **48**. In aspects, the optical assembly **50** may be supported in the recess **44** (FIG. 6) defined by the optical window **42** such that the optical assembly **50** is oriented on a sagittal plane of the user's oral cavity and towards buccal gingiva of the user when the oral appliance **10** is worn by the user.

(55) With reference to FIGS. 7 and 9, the optical housing **56** may include a plurality of isolation baffles or barriers **58** coupled to the printed circuit board **48**. The barriers **58** are arranged in parallel relation with one another. Adjacent barriers, such as, for example, barriers **58a**, **58b**, define a first space **62** therebetween in which one or more components of the oxygen sensor **60** are positioned. The oxygen sensor **60** may be a reflectance pulse oximeter configured to monitor/sense the oxygen saturation of a user by analyzing the change in color of the user's blood. The reflectance pulse oximeter **60** may measure the pulse rate of the user, typically in beats per minute, based on variations and/or deviations in the user's oxygen saturation. An exemplary pulse oximeter may use light-based technology to sense the actual oxygen saturation of hemoglobin of the user. According to an aspect, the pulse oximeter **60** includes light emitting diodes (LEDs) **60a** and photodiodes (PD) **60b**. The LED **60a** is configured to transmit red and infrared lights to vascular surfaces of the user's gingiva "G" and the photodiodes **60b** receive the light reflected from the gingiva "G" to determine changes in oxygen level in the user's gingiva "G" (e.g., the alveolar mucosa). In an aspect, the transmitted light may include blue light or green light, or a combination of lights according to the exemplary embodiments or otherwise consistent with this disclosure.

(56) The barriers **58** provide optical isolation between the LED **60a** and the detection photodiode **60b** to improve a signal-to noise ratio through tissue. For example, as shown in FIG. 9, the LED **60a** may be received in the first space **62** between adjacent barriers **58a**, **58b**, and the photodiode **60b** may be received in a second space **64** defined between adjacent barriers **58b**, **58c**. Each of the isolation barriers **58** may include a horizontal wall **66** extending perpendicularly therefrom to prevent laterally-projecting light from escaping the spaces **62**, **64** between the barriers **58**. In this way, the barriers **58** are configured to isolate the light projecting from the LEDs **60a**. Moreover, the optical reflections reject both dc and ac ambient light interference while directly improving signal quality and effectively reducing errors in physiological data acquisition. The spacing between the LEDs **60a** and the photodiodes **60b** is optimized to reduce backscatter from "blood-less" tissues and the barriers **58** reduce LED **60a** to photodiode **60b** crosstalk.

(57) Current commercial FDA cleared pulse oximeter devices are used, for example, to measure a body's oxygen saturation by using optical modules. These devices are typically used externally on the skin surface and measure oxygen saturation via transmissive or reflectance technique(s) using one LED and one photo diode. According to the FDA, SpO₂ readings obtained by these devices should be considered an estimate of arterial oxygen saturation. For example, commercial medical grade oximeters may record oxygen saturation values which are between about 4% and about 6% of the actual arterial blood oxygen readings. For example, if an FDA-cleared pulse oximeter reads 90%, then the true oxygen saturation in the blood is generally between 86-94% or possibly 84-96%.

(58) Oral mucosa in particular comprises a different tissue histology than skin, therefore the optical module design of existing commercial pulse oximeter devices is not necessarily optimal for intra-oral oxygen saturation data collection. At least one possible reason an existing commercial pulse oximeter device may not be optimal for intra-oral use is the number of LEDs and PDs as well as their positions relative to one another. In intra-oral applications, the light intensity and the angle of reflection of the light against the tissue may cause saturation (too much light intensity) or scattering of the light. Oral mucosa requires less depth penetration for the light and is prone to having excessive light scattering of the glossy mucosa layer, secondary to the high moisture content of the intra-oral tissue. The epithelial thickness of the oral mucosa is roughly between 0.3 mm-0.5 mm whereas the epithelial thickness of the skin, i.e., fingertip is between about 1 mm-3 mm.

(59) FIG. 9A shows an exemplary embodiment of an intra-oral oxygen sensor configured for minimizing the effect of light scatter from the oral mucosa according to this disclosure. The oxygen sensor **60** according to the exemplary embodiment is positioned on optical assembly **50** and includes without limitation an LED **60a** and two or more PDs **60b**. The oxygen sensor **60** is dimensioned and designed to position the two PDs **60b** on opposite sides of the LED **60a** at respective distances d , d' (for example, centerline to centerline) from the LED **60a**. In an exemplary configuration, each of the distances d , d' is in the range of approximately 3 mm to approximately 5 mm for optimal intra-oral measurement depths up to approximately 2 mm. The exemplary configuration has been found to result in a higher yield in PPG wave quality with less scattering. This effect may be due at least in part to the fact that the oral mucosa is thinner than the skin of the finger and less depth is needed to reach the vascular complexes at which the measurements will be performed within the oral tissue. The exemplary embodiment and configuration shown and described with respect to FIG. 9A has also been found to enhance the quality of the physiological data collection with lower LED power consumption, since the mucosal thickness does not require a deeper light penetration. The exemplary configuration has also been found to increase the speed at which the sensor reacts to changes in blood oxygen levels.

(60) In other configurations the distances d , d' may be varied and different configurations have been found to have optimal sensitivity within particular depth ranges. For example, distances within a range of about 6 mm to 8 mm had peak sensitivity at about 2 mm depth, distances within about 9 mm to about 14 mm had peak sensitivity at about 3 mm to 4 mm depth, and distances on the order of about 20 mm to about 30 mm had peak sensitivity at about 5 mm to about 6 mm depth.

(61) In an aspect and without limitation, a biocompatible adhesive or cohesive material may be applied to coat and hermetically seal over the optical components (e.g., LEDs/PDs) in any of the exemplary embodiments according to this disclosure or embodiments consistent with this disclosure. The biocompatible adhesive or cohesive material may have a low refraction index. In another aspect, the biocompatible adhesive or cohesive material may be less than about 1 mm thick. The biocompatible adhesive or cohesive material may be used in addition to or in substitution to the optical housing **56**.

(62) In aspects, the oral appliance **10** may further include one or more of the following additional components: a pressure sensor (not labeled), an airflow sensor (not labeled), a noise detector (not labeled), an actigraphy sensor (not labeled), a stimulator (not labeled), and an electroencephalogram (EEG) sensor **70** (FIG. 4) configured to differentiate various EEG rhythms and to determine, based on these detected rhythms, the various consciousness and sleep states of the user. It is contemplated that the oral appliance **10** may be able to assist with the diagnosis of brain related health conditions, such as epilepsy, by way of recording the user's brain activity-such brain activity may be collected by a data recorder (not labeled) for review and analysis. For further details regarding the EEG sensor **70**, reference may be made to U.S. patent application Ser. No. 16/781,417 filed Feb. 4, 2020 (now U.S. Pat. No. 11,375,951), the entire contents of which are incorporated by reference herein. The EEG sensor **70** may be positioned at a portion of the mouthpiece **20** that is adjacent a buccal side of a maxillary bone of the user when the oral appliance

is worn. That is, the EEG sensor may be located over the alveolar mucosa between the upper gums and inner lip/cheek of the user.

(63) According to an aspect, the oral appliance **10** may include a transceiver (not labeled). The transceiver may be configured to remotely monitor any additional components provided on and/or within the mouthpiece **20**. In an embodiment, the transceiver may be configured for use with a customized web-based application for a handheld wireless communication device. The customized web-based application may include features such as, a graph of the user's sleep position and chart and/or graphical data related to oxygen saturations of hemoglobin and the pressure applied to occlusal surfaces of the user's dentition. According to an aspect, the customized web-based application may include data related to the user's heart rate. In an embodiment, the transceiver communicates with handheld wireless communication devices having Bluetooth® capabilities. The transceiver may be communicable with handheld wireless communication devices, such as, for example, computers, smart watches, smart phones, and the like.

(64) In an embodiment, and with particular reference to FIGS. **10-14**, an oral appliance **100** for treatment of sleep apnea in a user is provided. The oral appliance **100** generally includes an upper mouthpiece **102** for positioning over upper dentition of a user, a lower mouthpiece **104** for positioning over lower dentition of the user, an electronic assembly **134** coupled to the upper mouthpiece **102**, and a pair of bilateral struts **108, 110** for coupling the upper and lower mouthpieces **102, 104** together. Each of the upper mouthpiece **102** and the lower mouthpiece **104** includes a socket **112, 114** coupled to opposing outer walls thereof. In aspects, the sockets **112, 114** may be monolithically formed with the upper mouthpiece **102** and the lower mouthpiece **104**.

(65) Each of the struts **108, 110** includes an elongate first portion **116** having a first projection **118** configured for receipt in the socket **112** of the upper mouthpiece **102**, and an elongate second portion **120** extending perpendicularly from the elongate first portion **116** and having a second projection **122** configured for receipt in the socket **114** of the lower mouthpiece **104s**. The oral appliance **10**, in combination with the struts **108, 110**, may be configured to position the bottom jaw of a user in a direction that is forward from the natural position of the user's jaw to aid in the treatment of sleep apnea in the user. The best jaw position for the user may be determined by a dentist and prescribed to a lab that is tasked with fabricating the oral appliance **10**. Further details regarding the sockets **112, 114** and the struts **108, 110** may be found in U.S. patent application Ser. No. 17/737,470 filed May 5, 2022, the entire contents of which being incorporated by reference above.

(66) The upper mouthpiece **102** includes a housing **130**, a cover **132**, and an electronic assembly **134** coupled to the housing **130**. The housing **130** has a first portion or occlusal portion **130a** (e.g., a top portion) configured to engage an occlusal surface of a user's dentition, and an opposite second portion **130b** (e.g., a bottom portion) defining an inner chamber **136**. The inner chamber **136** may extend along the entire arcuate length of the upper mouthpiece **102**. The second portion **130b** of the housing **130** has an outer peripheral edge **138** defining a groove **140**, and the cover **132** has an outer peripheral edge or ridge **144** configured for complimentary engagement with the groove **140**. The cover **132** of the housing **130** is secured to the second portion **130b** of the housing **130** to seal the inner chamber **136** from an external environment. It is contemplated that a sealing mechanism may be provided between at least a portion of the outer peripheral edge or ridge **144** and the groove **140**. A biocompatible adhesive may be provided between the groove **140** and ridge **144** to secure the cover **132** and the housing **130** to one another. It is contemplated that the biocompatible adhesive may have sealing capabilities.

(67) The cover **132** has a pair of posterior ends **132a, 132b** (FIG. **13**) each defining a port **146a, 146b** therethrough in communication with the inner chamber **136**. In aspects, the second portion **130b** of the housing **130** may define ports in posterior ends thereof. The ports **146a, 146b** may each have a filter or impermeable mesh cover to prevent fluids or debris from entering the inner chamber **136** while permitting sound to pass therethrough. The upper mouthpiece **102** may include a pair of

microphones **148**, such as, for example, Micro-Electro-Mechanical Systems (“MEMS”) microphones (e.g., the INFINEON MEMS mic), positioned in the inner chamber **136** adjacent the respective ports **146a**, **146b**. Alternatively, the microphone **148** may be positioned outside of the housing **130**. The microphone **148** may be covered and sealed with a biocompatible semi-permeable membrane (e.g., Gore-Tex® waterproof, breathable membrane) to keep moisture and particles away from the MEMS microphone **148**. The microphones **148** are configured to detect at least one of the user's normal breathing, changes in the user's breath sound intensity, or the user's snore sound quality. The microphones **148** may be configured to detect upper airway collapse in instances of obstructive sleep apnea. It is contemplated that the microphones **148** may be configured to identify specific breathing patterns associated with sleep-disordered breathing. A printed circuit board **150** may be provided in the inner chamber **136** in addition to other electronic components, including one or more of a pressure sensor, an airflow sensor, a noise detector, an actigraphy sensor, a stimulator, a oxygen saturation (SpO₂) sensor, a combined gyroscope and accelerometer, or an electroencephalogram (EEG) sensor.

(68) The microphones **148** may be pre-amplified and configured to record or sample sound at a frequency of about 10 kHz (thereby digitizing the analog sound signal captured). According to an aspect, the microphones **148** are configured to filter signals with: high pass $\tau=0.004$ s [order 2]; and low pass $\tau=0.0005$ s [order 2], wherein a root-mean-squared (RMS) signal (moving median $\frac{1}{6}$ th) is calculated to capture the magnitude of the sound. RMS computes the square root of the mean square of the instantaneous values of the signal. This processing, when applied to each sample of a continuous signal, simply makes all negative values of a signal positive, while keeping positive values as positive. Subsequently, a moving time average may be performed to describe the changes in the RMS signal magnitude over time (i.e. within a breath). Specifically, a moving median may be used, which may describe the median value over a 0.33-second window (window half-width $w=0.167$ seconds; the output signal at time t is the median value of the input data points from time $t-w$ to $t+w$ seconds).

(69) In at least one configuration, the microphones **148** include a S-VM3000-c Vesper-Mouser.

(70) According to an aspect, the upper mouthpiece **102** includes a processor operably coupled to the microphones **148**. The processor in combination with the microphones **148** are configured to identify the user's total sleep time, and total number of sleep disturbances followed by arousals during respiratory events. The microphones **148** are configured to detect changes in the user's air exchange intensity, and the processor is configured to translate the detected changes and trigger a stimulator to stimulate a muscle in the user's oral cavity to reverse the obstruction of the upper airway.

(71) The microphones **148** in combination with the processor are configured to detect at least one of snoring and awake periods or arousal during the user's sleep.

(72) According to an aspect, the upper mouthpiece **102** further includes a storage chip configured to store and record data; and a battery configured to provide power to at least one of the sound measuring device, the processor, and the storage chip. According to an aspect, the data can also be lived streamed and stored externally, if desired by the operator.

(73) As shown in FIGS. **10**, **13**, and **14**, the electronic assembly **134** may be external to the upper mouthpiece **102** and may extend upwardly from the upper mouthpiece **102**. In aspects, the electronic assembly **134** may be encased in a transparent outer housing (not explicitly shown). The electronic assembly **134** includes a substrate **152**, an oxygen sensor **154**, a printed circuit board **156**, an optical housing (not labeled) supported on the printed circuit board **156**, and a charging coil **158**. The substrate **152** has a first end portion **152a** coupled to the housing **130**, and a second end portion **152b** extending away from the housing **130** in a direction toward gingiva of a user. The first end portion **152a** of the substrate **152** is coupled to the housing **130** of the upper mouthpiece **102** and the second end portion **152b** of the substrate **152** supports the printed circuit board **156** thereon. The optical housing is supported on the printed circuit board **156** and the oxygen sensor **154** is

supported in the optical housing and electrically coupled to the printed circuit board **156**. The optical housing may be substantially similar to or identical to the optical housing **56** (FIG. 7) described above.

(74) The oxygen sensor **154** is configured to be oriented toward alveolar mucosa or gingiva of the user to determine an oxygen level of the user. The oxygen sensor **154** may be a reflectance pulse oximeter configured to monitor/sense the oxygen saturation of a user by analyzing the change in color of the user's blood. The reflectance pulse oximeter **154** may measure the pulse rate of the user, typically in beats per minute, based on variations and/or deviations in the user's oxygen saturation. An exemplary pulse oximeter may use light-based technology to sense the actual oxygen saturations of hemoglobin of the user. According to an aspect, the pulse oximeter **154** may be substantially similar to or identical to the pulse oximeter **60** described above.

(75) FIGS. **15-23** illustrate charts that depict the results of sounds that were measuring using a sound measuring device, according to an aspect. The data illustrated in the charts of FIGS. **15-23** was collected at different time intervals of a snoring test utilizing a sound measuring device (e.g., a microphone) of an oral appliance of the present disclosure. The sound measuring device utilized was a microphone embedded in the oral appliance. The sound measuring device utilized was positioned in a sealed circular housing that was open to one side and coupled to an opposing side by an adhesive. The sound measuring device was positioned at a location inferior to one of the user's upper rear molar.

(76) The objective of the tests was to determine whether loud snoring sounds and quiet breathing sounds, as seen at the trachea, would be clearly detectable in the user's mouth with this oral device. The subject, awake, performed a series of inspiratory snoring sounds followed by periods of normal, quiet breathing. The oral microphone signals were captured well-snoring sounds were detected without clipping and breathing sounds were detected above the noise floor.

(77) FIG. **15** is a chart that depicts data relating to snoring and normal breathing sounds, collected by a sound measuring device. As illustrated in the chart, sound was measured starting from about 498 seconds into the user's sleep cycle. The user made snoring sounds on inspiration and breathed normally on expiration. The dashed lines around the 512-516 second marks validates the data collected from the sound measuring device given that the data collected by the sound measuring device of the oral appliance is in line with the illustrated standard reference data from collected from an external (e.g., tracheal) microphone.

(78) FIG. **16** is a chart that depicts data relating to snoring and normal breathing sounds, collected by a sound measuring device, according to an aspect. As illustrated in the chart, sound was measured from about 512 seconds into the user's sleep cycle. The user made snoring sounds on inspiration and breathed normally on expiration.

(79) FIG. **17** is a chart that depicts data relating to normal breathing sound, collected by a sound measuring device, according to an aspect. As illustrated in the chart, sound was measured from about 230 seconds into the user's sleep cycle. The user breathed normally during that time period without making any sounds.

(80) FIG. **18** is a chart that depicts data relating to normal breathing sound, collected by a sound measuring device, according to an aspect. As illustrated in the chart, sound was measured from about 572 seconds into the user's sleep cycle. The user breathed normally during that time period.

(81) FIG. **19** is a chart that depicts data relating to snoring and normal breathing sounds, collected by a sound measuring device, according to an aspect. As illustrated in the chart, the user made snoring sounds then breathed normally.

(82) FIG. **20** is a chart that depicts data relating to normal breathing sound, collected by a sound measuring device, according to an aspect. As illustrated in the chart, sound was measured from about 533 seconds into the user's sleep cycle. The user breathed normally during that time period.

(83) FIG. **21** is a chart that depicts data relating to snoring sounds, collected by a sound measuring device, according to an aspect. The user made snoring sounds during the test.

(84) FIG. 22 is a chart that depicts data relating to snoring sounds, collected by a sound measuring device, according to an aspect. As illustrated in the chart, the user made snoring sounds during the test.

(85) FIG. 23 is a chart that depicts data relating to normal breathing sound, collected by a sound measuring device with a signal amplitude dropped due to saliva, according to an aspect. As illustrated in the chart, the user breathed normally during the test.

(86) In aspects, the present disclosure may provide an oral appliance that includes an inertial measuring unit (IMU) (e.g., the BOSCH 6 axis accelerometer, gyroscope) configured to measure triaxial acceleration and triaxial angular velocity. According to an aspect, the IMU includes a gyroscope and an accelerometer. The gyroscope is configured to output angular velocity signals on three axes in space, while the accelerometer is configured to output linear acceleration signals on three axes in space. According to an aspect, the IMU is a miniature IMU sized so that it can be coupled to the mouthpiece. The miniature IMU may be secured to a buccal, lingual, or occlusal wall of the mouthpiece. The placement of the IMU or miniature IMU may be at the most anterior segment of the user's lower jaw (that is, the location of the user's lower front incisors). It is contemplated that this may be an optimal location on the mouthpiece at least because this anatomical location will likely express the most detailed movement during recording. This follows the physics law of "the fulcrum effect". The closer the point of effort is to the fulcrum (the back of the jaw where the joints and associated muscles and ligaments are), the less effort it will take for the movement at the farthest point to the fulcrum (most anterior part of the jaw)-hence the more sensitive that point is to detecting movement.

(87) It is contemplated that the oral appliance may be configured for the detection of lower jaw movements. These lower jaw movements may serve as a surrogate bio-signal for the detection of breathing effort, including snoring and arousal during sleep. According to an aspect, the lower jaw movements may be in addition to other bio-signals that are indicative of the presence of sleep apnea in a user. Jaw movement bio-signals can identify specific breathing patterns associated with sleep-disordered breathing, including obstructive sleep apnea ("OSA") The use of the oral appliance including the IMU or miniature IMU can indicate sleep vs awake stage, and can determine OSA severity levels in those patients. Further analysis using specific algorithms can also identify total sleep time, total number of sleep disturbances followed by arousals during respiratory events or OSA events. According to an aspect, the algorithm may also be used as a predictive model for future systemic issues such as cardiovascular disease (CVD). It is contemplated that the IMU or accelerometer-gyroscope combination may determine jaw clenching and grinding which is an indicator for arousals during periods of disturbed sleep. The IMU may also be an indicator of treatment compliance for a patient undergoing treatment for OSA.

(88) FIGS. 24A-26B include data collected by an IMU as described above.

(89) FIGS. 24A and 24B compare data collected by an IMU coupled to an oral appliance of the present disclosure with data collected by an IMU extra-orally coupled to the patient's chin, where a patient is opening and closing their mouth and their head is in a straight supine position. This data may be important in determining whether the patient is a "mouth breather" during sleep and the extent of their mouth breathing.

(90) FIGS. 25A and 25B compare data collected by an intra-oral IMU coupled to an oral appliance of the present disclosure with data collected by the IMU extra-orally coupled to the patient's chin, where a patient is grinding their teeth in a side-to-side motion and their head is in a straight supine position. This data may provide an indication of sleep bruxism secondary to obstructive sleep apnea.

(91) FIGS. 26A and 26B compare data collected by an IMU coupled to an oral appliance of the present disclosure with data collected by the IMU extra-orally coupled to the patient's chin, with a patient swallowing.

(92) Swallowing is a strong indicator of arousal during sleep, most frequently occurring in REM.

The intra-oral IMU of the present disclosure has a very distinct and consistent wave shape during swallowing, compared to the external chin sensor.

(93) With reference to FIGS. 27-30, another embodiment of an oral appliance **300** is provided, similar to the oral appliances **10**, **100**, **200** described above. The oral appliance **300** generally includes a mouthpiece **302** and a sensor assembly **350** positioned in an appendage or box **304** extending perpendicularly away from a posterior end portion of the mouthpiece **302**. The sensor assembly **350** may be positioned adjacent an optical window **342** (FIG. 30) of the appendage **304** and includes a charging coil **354**, a printed circuit board **356**, a spacer **358**, and an oxygen sensor **360**. The oxygen sensor **360** may be oriented on a sagittal plane of the user's oral cavity and towards buccal gingiva of the user when the oral appliance **300** is worn by the user. The oxygen sensor **360** may be similar to any of the oxygen sensors described above.

(94) The charging coil **354** is stacked over the printed circuit board **356** and separated from the printed circuit board **356** by the spacer **358** to reduce heat that is produced by the charging coil **354** during charging. As such, the spacer **358** keeps excess heat away from the printed circuit board **356**. The appendage **304** is specially designed to allow for the charging coil **354** to be housed in a recessed part **306** of the appendage **304**, which is about 0.6 mm in thickness relative to the rest of appendage **304**. This minimal thickness (e.g., about 0.6 mm) will allow for more efficient contact between the charger (not explicitly shown) and the charging coil **354**. The appendage **304** may have another recessed part **362** (FIG. 30) with a minimal thickness relative to the rest of the wall of the appendage **304** to allow for better transmission of the PD and LED of the oxygen sensor **360**.

(95) With reference to FIGS. 31-33, the present disclosure provides another embodiment of an oral appliance **400** for treating and/or collecting data used for treating sleep apnea. During sleep, electromyogram (EMG) can serve as an indicator of atonia (temporary paralysis of the user's legs and arms) while in REM vs non-REM sleep. It is contemplated that the oral appliance **400**, which includes an EMG sensor, can provide data that would be indicative of respiratory events and arousals from apnea, and bruxism during sleep disturbances. EMG activity in subjects with sleep disorders may be typically more than twice as many as those without sleep disorders. It is contemplated that measuring the EMG of a user's intra-oral muscles may provide more sensitive and accurate data sets compared to external muscle EMG such as on the chin or the tibia.

(96) The tongue is the largest and most active muscle in the oral cavity. It is also the upper respiratory muscle which is most involved in controlling air flow to the back of the throat during sleep. Hence, detecting its contractility during sleep can provide valuable physiological data for monitoring and diagnostic purposes. Because of the tongue's important role in respiration, detecting its electric activity during obstructive sleep apnea is valuable. The contemplated oral appliance **400** may be configured to detect such activities by placing EMG sensors along the right and left sides of the inferior-lateral aspects of the tongue while coupled to a removable mouthpiece. It is contemplated that the EMG sensor may partially or fully cover the anterior and posterior of the user's tongue, detecting the expiratory and inspiratory motor units associated with muscle fibers of the user's tongue.

(97) As illustrated FIG. 31, an exemplary oral appliance **400** includes a mouthpiece **402** configured for being positioned in an oral cavity of a user. According to an aspect, the mouthpiece **402** includes a lingual wall **404** configured for being placed adjacent the user's tongue. The mouthpiece **402** may be made according to any of the methods described above and from any of the materials described above. The mouthpiece **402** is configured to be positioned on the user's lower jaw or upper jaw.

(98) The oral appliance **400** further includes a pair of electromyogram (EMG) EMG sensors **406**, **408** (e.g., the MYOWARE muscle sensor). The EMG sensors **406**, **408** are configured to provide physiological data for monitoring and diagnosing obstructive sleep apnea. According to an aspect, the EMG sensors **406**, **408** are configured to detect an electrical activity of the user's tongue. The electrical activity may be indicative of a muscle contraction of the user's tongue. The EMG sensors

406, 408 are configured to be in contact with right and left sides of the user's tongue such that the EMG sensors **406, 408** detect twitches or contractions of the user's tongue bilaterally. The EMG sensors **406, 408** may be disposed on the mouthpiece **402** such that the EMG sensors **406, 408** contact the right and left sides of the inferior-lateral aspects of the user's tongue. It is contemplated that the EMG sensors **406, 408** may be sized so that they contact a partial area of the side(s) of the user's tongue. According to an aspect, the EMG sensors **406, 408** may be sized so that they can fully cover the anterior and posterior lateral surfaces of the user's tongue. It is contemplated that by being positioned to be in contact with the lateral surfaces of the user's tongue, the EMG sensors **406, 408** detect the expiratory and inspiratory motor units associated with muscle fibers of the user's tongue.

(99) According to an aspect, the oral appliance **400** may include a single EMG sensor that contacts a single lateral surface of the user's tongue. Alternatively, multiple EMG sensors may be disposed on a single side of the mouthpiece **400** so that they all contact the same side (right or left lateral sides, for example) of the user's tongue. While FIG. 31 illustrates that the oral appliance **400** may include a wire **410** extending from the EMG sensors **406, 408**, it is contemplated that the oral appliance **400** may be wireless.

(100) According to an aspect, the oral appliance **400** may include a processor (not explicitly shown). The processor may include a micro-processor. According to an aspect, the EMG sensors **406, 408**, in combination with the micro-processor, may be configured to identify respiratory events and arousals from obstructive sleep apnea. The EMG sensors **406, 408**, in combination with the micro-processor, may be configured to identify bruxism during sleep disturbances.

(101) The oral appliance **400** may further include a storage chip (not explicitly shown) configured to store and record data collected by the EMG sensors **406, 408**. According to an aspect, a battery (not explicitly shown) may be provided to supply power to at least one of the EMG sensors **406, 408**, the processor or micro-processor, and the storage chip. The battery may be a rechargeable battery. The oral appliance **400** may be positioned on a charger platform (not explicitly shown) so that the rechargeable battery can be charged so that it can supply power to the oral appliance **400**.

(102) When worn during sleep, the oral appliance **400** may provide information that serves as an indicator of at least one of atonia while the user is in REM sleep, respiratory events and arousals from apnea, and bruxism during sleep disturbances.

(103) It is further contemplated that the oral appliance **400**, in addition to the EMG sensor(s) **406, 408**, may include an additional sensor (not explicitly shown). The additional sensor may be at least one of an oxygen sensor, a PPG sensor, or an inertial movement sensor. The EMG sensors **406, 408** and the additional sensor(s) may collectively gather physiologic data over time that aids in diagnosing obstructive sleep apnea. The micro-processor and rechargeable battery(s) may be sealed within the mouthpiece **400** or otherwise coupled to the mouthpiece **400**.

(104) Information or data gathered by the EMG sensors **406, 408** and the additional sensor(s) may be live streamed onto a smart device, for example, wirelessly via i.e., Bluetooth. The data can also be downloaded the next day by placing the appliance on a stationary hub/charger combination (not shown). The data can then be transferred onto a secure cloud based medium to be shared with the user as well as their care provider(s). The accumulation of this collective data may be built into a predictive model through deep thinking/artificial intelligence (AI) system integration to optimize the accurate diagnosis and subsequent monitoring of the ongoing treatment of patients suffering from sleep disordered breathing, i.e., obstructive sleep apnea. Additionally, the predictive model can assist in the determination of the most statistically efficacious prescribed therapy for OSA, with the highest probability of success, prior to the user engaging in any therapy which might provide sub-standard treatment response, such as positive air pressure (i.e., CPAP), custom oral sleep appliances, hypoglossal nerve stimulator, and pharmacotherapy.

(105) It is contemplated that the EMG sensors **406, 408** may be suitably coupled to any of the oral appliances described herein.

(106) The additional sensor coupled to the oral appliance 400 may be an EEG sensor. With reference to FIG. 32, EMG data was obtained using an EEG sensor to determine when a user's eyes are opened or closed: (Phase 1) the user's eyes were opened and then closed; (Phase 2) when the user's eyes were open; and (Phase 3) again, when the user's eyes were opened and then closed. During that test period, EMG data was gathered during Phase 1 and Phase 3. Distinct signals were obtained that reflected the electrical activity in the user's tongue.

(107) FIG. 33 is a chart that illustrates the difference between EMG data representing tongue contraction and EMG data representing an absence of tongue contraction. As illustrated, distinct signals are obtained that reflect when there is muscle contraction or twitching of the user's tongue, which indicates the user is not in REM sleep whereas signals reflecting tongue muscle relaxation indicates the user is in REM sleep. No distinct signals were created during the absence of muscle contraction or twitching of the user's tongue.

(108) The components of the apparatus illustrated are not limited to the specific embodiments described herein, but rather, features illustrated or described as part of one embodiment can be used on or in conjunction with other embodiments to yield yet a further embodiment. It is intended that the apparatus include such modifications and variations. Further, steps described in the method may be utilized independently and separately from other steps described herein.

(109) While the apparatus and method have been described with reference to specific embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope contemplated. In addition, many modifications may be made to adapt a particular situation or material to the teachings found herein without departing from the essential scope thereof.

(110) In this specification and the claims that follow, reference will be made to a number of terms that have the following meanings. The singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Furthermore, references to “one embodiment”, “some embodiments”, “an embodiment” and the like are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term such as “about” is not to be limited to the precise value specified. In some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Terms such as “first,” “second,” “upper,” “lower” etc. are used to identify one element from another, and unless otherwise specified are not meant to refer to a particular order or number of elements.

(111) As used herein, the terms “may” and “may be” indicate a possibility of an occurrence within a set of circumstances; a possession of a specified property, characteristic or function; and/or qualify another verb by expressing one or more of an ability, capability, or possibility associated with the qualified verb. Accordingly, usage of “may” and “may be” indicates that a modified term is apparently appropriate, capable, or suitable for an indicated capacity, function, or usage, while taking into account that in some circumstances the modified term may sometimes not be appropriate, capable, or suitable. For example, in some circumstances an event or capacity can be expected, while in other circumstances the event or capacity cannot occur—this distinction is captured by the terms “may” and “may be.”

(112) As used in the claims, the word “comprises” and its grammatical variants logically also subtend and include phrases of varying and differing extent such as for example, but not limited thereto, “consisting essentially of” and “consisting of.” Where necessary, ranges have been supplied, and those ranges are inclusive of all sub-ranges therebetween. It is to be expected that variations in these ranges will suggest themselves to a practitioner having ordinary skill in the art and, where not already dedicated to the public, the appended claims should cover those variations.

(113) Advances in science and technology may make equivalents and substitutions possible that are

not now contemplated by reason of the imprecision of language; these variations should be covered by the appended claims. This written description uses examples to disclose the method, machine and computer-readable medium, including the best mode, and also to enable any person of ordinary skill in the art to practice these, including making and using any devices or systems and performing any incorporated methods. The patentable scope thereof is defined by the claims, and may include other examples that occur to those of ordinary skill in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

Claims

1. An oral appliance, comprising: a mouthpiece configured to be positioned in an oral cavity of a user, the mouthpiece including: an anterior wall configured to cover facial surfaces of the user's dentition; a posterior wall configured to cover lingual surfaces of the user's dentition; a transverse wall extending between the anterior wall and the posterior wall; and an electronic assembly coupled to the mouthpiece, the electronic assembly including an oxygen sensor coupled to the anterior wall and configured to be oriented toward oral mucosa of the user's mouth to determine an oxygen saturation level of the user, the oxygen sensor comprising a light emitting diode (LED) and two or more photodiodes (PDs) in an arrangement, the LED configured to project both red and infrared lights, and each of the two or more PDs configured to detect reflections from the oral mucosa of at least a portion of both the red and infrared lights projected by the LED and generate photoplethysmography (PPG) data based on the reflections detected, wherein the oxygen sensor is configured to detect reflections from the oral mucosa via the arrangement, wherein in the arrangement, a first PD of the two or more PDs and a second PD of the two or more PDs are positioned on opposite sides of the LED, a first centerline of the first PD, a second centerline of the second PD, and a third centerline of the LED are positioned in parallel, the first PD has a major dimension extending along the first centerline and a minor dimension extending perpendicular to the first centerline such that the major dimension is larger than the minor dimension of the first PD, the second PD has a major dimension extending along the second centerline and a minor dimension extending perpendicular to the second centerline such that the major dimension is larger than the minor dimension of the second PD, and a distance between the first centerline of the first PD and the second centerline of the second PD is between 6 millimeters and 10 millimeters.
2. The oral appliance of claim 1, further comprising a housing positioned on the mouthpiece, wherein the housing includes, encompasses, or surrounds at least a portion of the electronic assembly.
3. The oral appliance of claim 1, wherein the electronic assembly further includes: a printed circuit board; and an optical housing supported on the printed circuit board, wherein the oxygen sensor is supported in the optical housing and electrically coupled to the printed circuit board.
4. The oral appliance of claim 1, wherein the oxygen sensor is a reflectance pulse oximeter configured to determine the oxygen saturation level of hemoglobin of the user via the reflections received by the two or more PDs.
5. The oral appliance of claim 1, wherein the electronic assembly has a first end portion coupled to the mouthpiece, and a second end portion configured to extend away from the mouthpiece in a direction toward gingiva of the user, the second end portion of the electronic assembly supporting the oxygen sensor thereon.
6. The oral appliance of claim 1, the electronic assembly including an electroencephalogram (EEG) electrode, wherein the EEG electrode is configured to be positioned adjacent the user's alveolar mucosa such that the EEG electrode detects brain wave signals.
7. The oral appliance of claim 1, further comprising a housing secured to the mouthpiece, opposite

the transverse wall, wherein the housing defines an inner chamber of the mouthpiece; and a microphone positioned in the inner chamber of the mouthpiece.

8. The oral appliance of claim 1, wherein a microphone is positioned on the mouthpiece and the microphone is covered by a semi-permeable membrane.

9. The oral appliance of claim 1, further wherein, the mouthpiece comprises a first mouthpiece and a second mouthpiece; the mouthpiece further comprising a strut including: an elongate first portion having a first projection configured for receipt in a socket of the first mouthpiece; and an elongate second portion extending perpendicularly from the elongate first portion and having a second projection configured for receipt in a socket of the second mouthpiece.

10. The oral appliance of claim 1, wherein the oxygen sensor is positioned above a top portion of the anterior wall.

11. The oral appliance of claim 1, further comprising an inertial measuring unit (IMU) coupled to the mouthpiece and configured to detect movement of the user.

12. The oral appliance of claim 1, wherein the electronic assembly includes a plurality of the oxygen sensors and each of the plurality of the oxygen sensors include a respective one of the LED and a respective two or more of the PDs.

13. An oral appliance, comprising: a mouthpiece configured to be positioned in an oral cavity of a user, the mouthpiece including: an anterior wall configured to cover facial surfaces of the user's dentition; a posterior wall configured to cover lingual surfaces of the user's dentition; a transverse wall extending between the anterior wall and the posterior wall; and an electronic assembly coupled to the mouthpiece, the electronic assembly including an oxygen sensor coupled to the anterior wall and configured to be oriented toward oral mucosa of the user's mouth to determine an oxygen saturation level of the user and collect oxygen saturation (SpO₂) and photoplethysmography (PPG) data, the oxygen sensor comprising a light emitting diode (LED) and two or more photodiodes (PDs) in an arrangement, the LED configured to project both red and infrared lights, and each of the two or more PDs configured to detect reflections from the oral mucosa of at least a portion of both the red and infrared lights projected by the LED, wherein the oxygen sensor is configured to detect reflections from the oral mucosa via the arrangement, wherein in the arrangement, a first PD of the two or more PDs and a second PD of the two or more PDs are positioned on opposite sides of the LED, a first centerline of the first PD, a second centerline of the second PD, and a third centerline of the LED are positioned in parallel, the first PD has a major dimension extending along the first centerline and a minor dimension extending perpendicular to the first centerline such that the major dimension is larger than the minor dimension of the first PD, the second PD has a major dimension extending along the second centerline and a minor dimension extending perpendicular to the second centerline such that the major dimension is larger than the minor dimension of the second PD, and a distance between the first centerline of the first PD and the second centerline of the second PD is between 6 millimeters and 10 millimeters.

14. The oral appliance of claim 13, further comprising an inertial measuring unit (IMU) coupled to the mouthpiece and configured to detect movement of the user.

15. The oral appliance of claim 13, wherein the electronic assembly has a first end portion coupled to the mouthpiece, and a second end portion configured to extend away from the mouthpiece in a direction toward gingiva of the user, the second end portion of the electronic assembly supporting the oxygen sensor thereon.

16. The oral appliance of claim 13, the electronic assembly including an electroencephalogram (EEG) electrode, wherein the EEG electrode is configured to be positioned adjacent the user's alveolar mucosa such that the EEG electrode detects brain wave signals.

17. An oral appliance, comprising: a mouthpiece configured to be positioned in an oral cavity of a user, the mouthpiece including: an anterior wall configured to cover facial surfaces of the user's dentition; a posterior wall configured to cover lingual surfaces of the user's dentition; a transverse

wall extending between the anterior wall and the posterior wall; an appendage configured to extend away from the anterior wall in a direction towards oral mucosa of the user's mouth and defining a space within the appendage; and an electronic assembly including an oxygen sensor and positioned at least partially within the space within the appendage, wherein the appendage and electronic assembly are together configured for orienting the oxygen sensor toward the oral mucosa of the user's mouth, the oxygen sensor comprising a light emitting diode (LED) and two or more photodiodes (PDs) in an arrangement, the LED configured to project both red and infrared lights, and each of the two or more PDs configured to detect reflections from the oral mucosa of at least a portion of both the red and infrared lights projected by the LED and generate photoplethysmography (PPG) data based on the reflections detected, wherein the oxygen sensor is configured to detect reflection from the oral mucosa via the arrangement, wherein in the arrangement, a first PD of the two or more PDs and a second PD of the two or more PDs are positioned on opposite sides of the LED, a first centerline of the first PD, a second centerline of the second PD, and a third centerline of the LED are positioned in parallel, the first PD has a major dimension extending along the first centerline and a minor dimension extending perpendicular to the first centerline such that the major dimension is larger than the minor dimension of the first PD, the second PD has a major dimension extending along the second centerline and a minor dimension extending perpendicular to the second centerline such that the major dimension is larger than the minor dimension of the second PD, and a distance between the first centerline of the first PD and the second centerline of the second PD is between 6 millimeters and 10 millimeters.

18. The oral appliance of claim 17, further comprising an inertial measuring unit (IMU) coupled to the mouthpiece and configured to detect movement of the user.

19. The oral appliance of claim 17, wherein the electronic assembly has a first end portion coupled to the mouthpiece, and a second end portion configured to extend away from the mouthpiece in a direction toward gingiva of the user, the second end portion of the electronic assembly supporting the oxygen sensor thereon.

20. The oral appliance of claim 17, the electronic assembly including an electroencephalogram (EEG) electrode, wherein the EEG electrode is configured to be positioned adjacent the user's alveolar mucosa such that the EEG electrode detects brain wave signals.
